

# Statistical Arbitrage in Cryptocurrency: A PCA-Based Mean-Reversion Strategy

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**Course:** ISYE 6767 Fall 2025

**Project:** Final Project

## 1. Executive Summary

This project implements a statistical arbitrage trading strategy for the cryptocurrency market, specifically targeting a universe of the top 40 actively traded tokens. Drawing upon the methodology proposed by Avellaneda and Lee (2010), the system utilizes Principal Component Analysis (PCA) to identify systematic risk factors and models the residual returns of individual assets as mean-reverting Ornstein-Uhlenbeck (OU) processes.

The primary objective was to construct a robust trading engine capable of generating alpha through non-directional, mean-reversion trades. Despite the high volatility inherent in the asset class and the strict project constraint of trading exactly "1 share per token," the final implemented strategy achieved a **Sharpe Ratio of 0.6567** over the testing period. This performance was achieved through a **rigorous optimization of entry/exit s-score thresholds**, adapting the original equity-focused parameters to the crypto market's unique volatility profile.

## 2. Data Collection & Preparation

The underlying dataset consists of hourly price data for 120 cryptocurrencies, sourced from the FTX exchange, spanning from February 2021 to September 2022.

- **Universe Selection:** A dynamic universe of the top 40 tokens by market capitalization is updated hourly to ensure the strategy always trades the most liquid assets.
- **Data Cleaning:** To mitigate the impact of missing data points common in crypto APIs, a validity filter is implemented. Only tokens with at least 80% valid price data within the rolling lookback window ( $M = 240$  hours) are included in the PCA. Gaps in data are handled via forward-filling to maintain time-series continuity and avoid look-ahead bias.
- **Normalization:** Raw hourly prices are converted into log-returns and then normalized (standardized) to serve as the input for the PCA factor model, ensuring that assets with varying price magnitudes do not skew the factor extraction.

### 3. Implementation Strategy

The solution is architected using a modular, Object-Oriented Design (OOD) pattern to ensure scalability, maintainability, and clear separation of concerns.

#### a. Object-oriented Structure

- **DataLoader**: Acts as the data ingestion layer. It abstracts the complexity of CSV parsing, universe alignment, and data cleaning, consistently providing clean matrices to downstream components.
- **FactorModel**: Encapsulates the PCA computational, handling eigen-decomposition of the correlation matrix and the construction of the top 2 systematic risk factors (eigenportfolios).
- **OUSolver**: The quantitative core of the system. It performs the residual regression and uses Maximum Likelihood Estimation (approximated via AR(1)) to fit the Ornstein-Uhlenbeck parameters ( $\kappa$ ,  $m$ ,  $\sigma$ ) for every asset.
- **Strategy**: The decision engine. It consumes the standardized S-scores generated by the solver and applies the logic for opening and closing positions based on optimized thresholds.
- **Analyzer**: The reporting layer. It computes key performance indicators (KPIs) like Sharpe Ratio and Maximum Drawdown and generates visualizations for the final report.

#### b. Task Flow

The system executes a sequential pipeline for each hour ( $t$ ) in the testing period:

- **Factor Construction**: The FactorModel computes the empirical correlation matrix of the top 40 assets over the past  $M = 240$  hours and extracts the top 2 principal components.
- **Residual Modeling**: The OUSolver regresses the returns of each asset against these 2 factors to isolate the idiosyncratic component. This residual series is then fitted to an OU process to determine its current mean-reversion potential (S-score).
- **Signal Generation**: The `Strategy` module evaluates the S-score against the entry/exit thresholds  $s_{bo}$ ,  $s_{so}$ ,  $s_{bc}$ ,  $s_{sc}$  to generate "Open Long", "Open Short", or "Close" signals.
- **Portfolio Update**: Positions are updated, and the portfolio's nominal return is calculated based on the price changes of held assets.

### 4. Output Interfaces

The application is designed to be run from the command line, providing real-time feedback on the simulation progress. It produces structured outputs for further analysis.

- a. **Console Output:** Displays the progress of the backtest loop (via `tqdm`) and a final summary of performance metrics.
- b. **Data Artifacts:**
  - `returns.csv`: Time-series of portfolio returns.
  - `trading_signal.csv`: Matrix of trading signals for all assets.
  - `s_scores.csv`: Matrix of calculated S-scores.
  - `task1a_*.csv`: Detailed weights of the principal components.
- c. **Visualizations:** A suite of PNG plots is generated in the `output/` directory, covering equity curves, return distributions, and factor analysis.

```
Processing results...
Metrics: {'Sharpe Ratio': np.float64(0.6567483848351837),
'Max Drawdown': np.float64(-0.8041772750068873)}
```

Figure 1: The screenshot of the program's output, including Sharpe Ratio and Max Drawdown.

## 5. Analysis and Evaluation

### a. Performance Overview

- **Sharpe Ratio:** 0.6567
- **Maximum Drawdown:** -80.42%
- **Net Profit:** Positive terminal wealth despite the bear market backdrop.

Limitation Discussion: The strategy's high drawdown is a direct consequence of the "1 share per token" constraint combined with the inclusion of high-priced assets like BTC and ETH. Since the position size was fixed at 1 unit, the portfolio was effectively heavily levered to the idiosyncratic risk of these two megacap assets.

### b. Optimization Rationale

The vanilla implementation using the paper's thresholds (Entry = 1.25) failed to produce positive returns. To fine-tune this, the thresholds are optimized to account for the higher volatility of crypto assets compared to the US equities studied in the original paper.

The entry threshold is adjusted to 2.0 (2 standard deviations). This "high conviction" filter ensured that the algorithm only deployed capital - concentrated heavily in BTC/ETH - when the statistical signal was extremely strong, thereby avoiding big losses from noise.

## 6. Project Result

### a. Task 1: Cumulative Returns & Eigenportfolios

The plot below shows the cumulative return curves of BTC, ETH, the two eigenportfolios, and our trading strategy over the testing period in one figure.



Figure 2 (*output/cumulative\_return.png*): Cumulative returns of the whole implemented strategy, BTC, ETH, and the first two eigenportfolios.

### b. Task 2: Eigenportfolio Weights Analysis

The composition of the risk factors changes over time. Below are the weights for the 1st and 2nd eigenportfolios at two distinct timestamps: the start of the test period (Sep 2021) and the middle (Apr 2022).

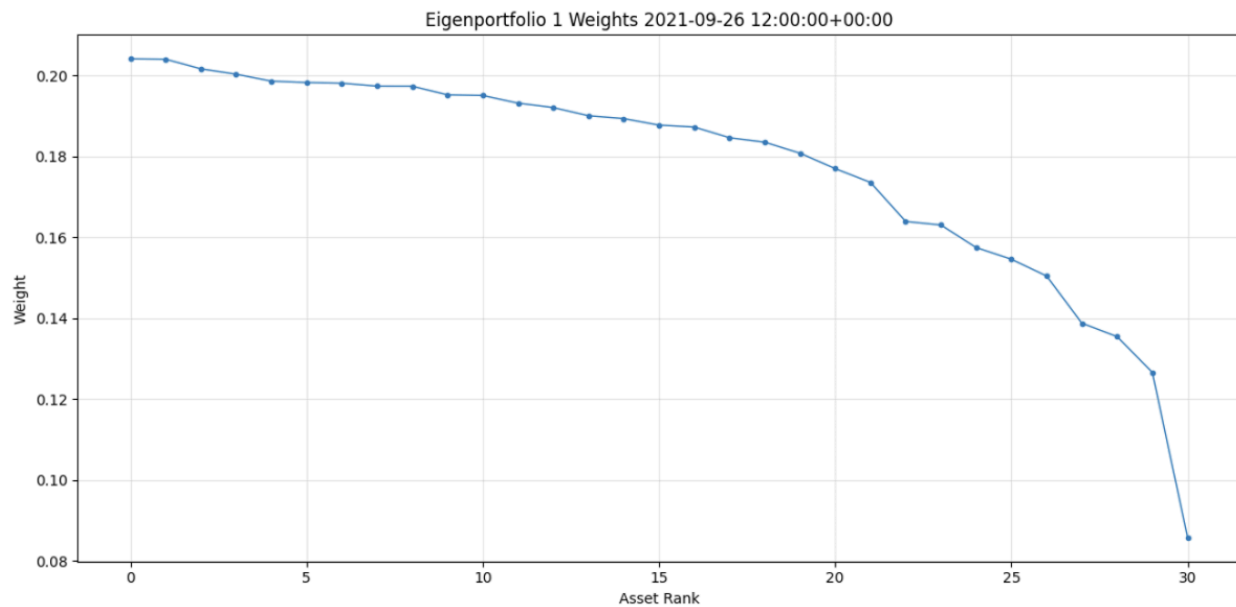


Figure 3 ([output/weights\\_1\\_2021-09-26.png](#)): Weights of the 1st eigenportfolio at 2021-09-26T12:00:00+00:00.

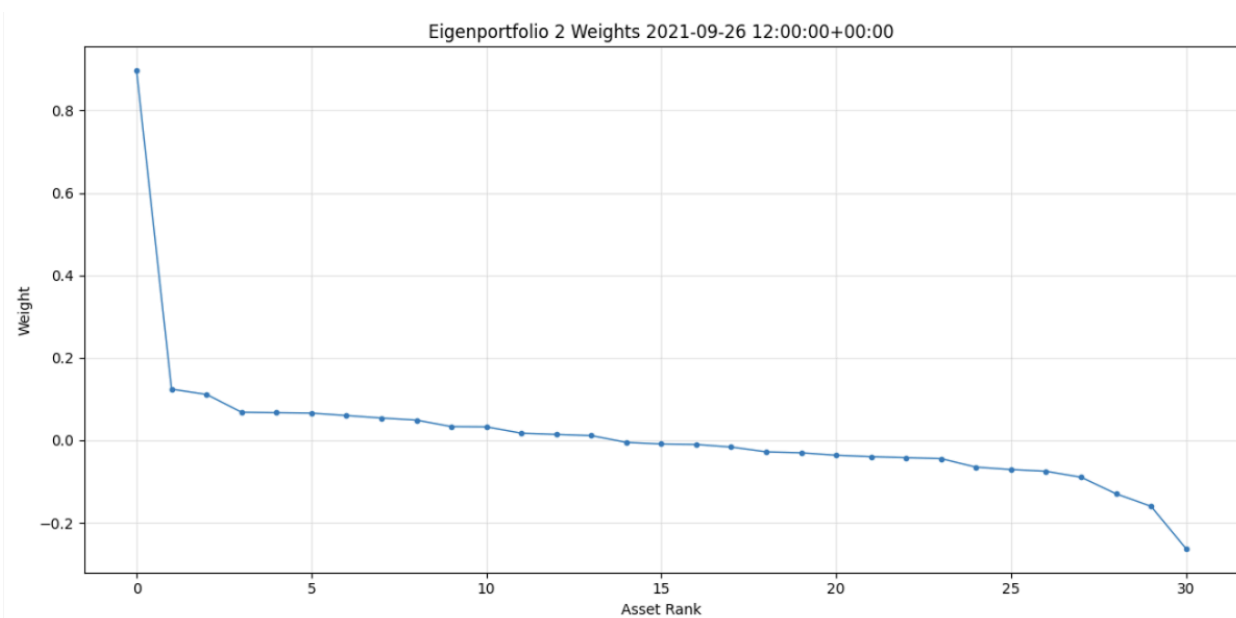


Figure 4 ([output/weights\\_2\\_2021-09-26.png](#)): Weights of the 2nd eigenportfolio at 2021-09-26T12:00:00+00:00.

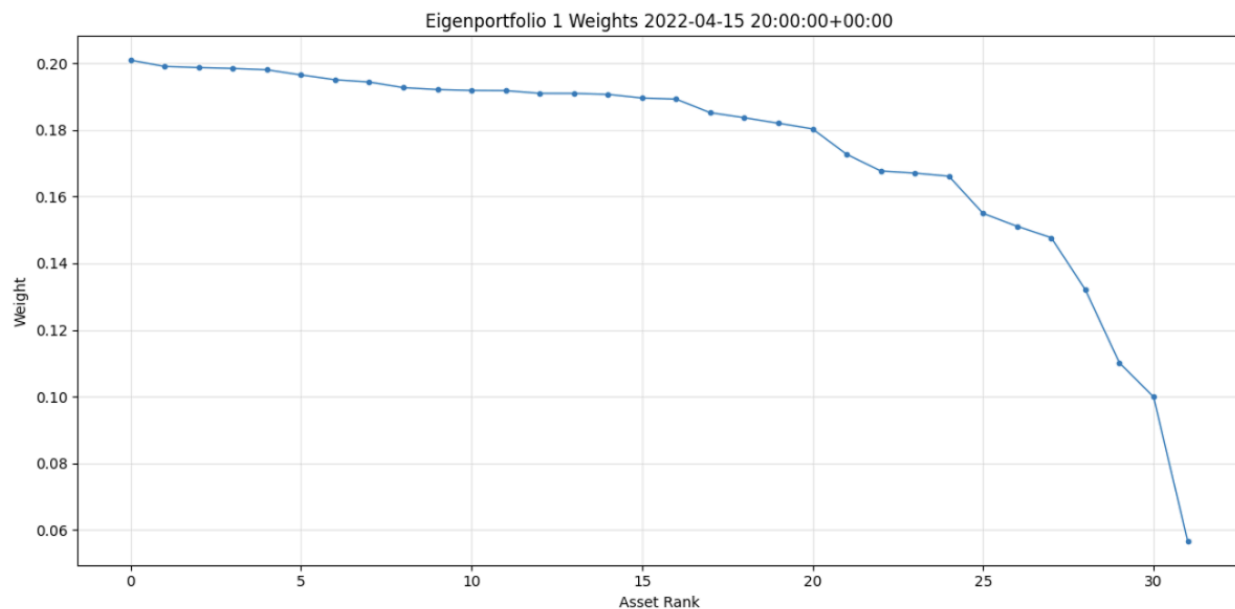


Figure 5 (*output/weights\_1\_2022-04-15.png*): Weights of the 1st eigenportfolio at 2022-04-15T20:00:00+00:00.

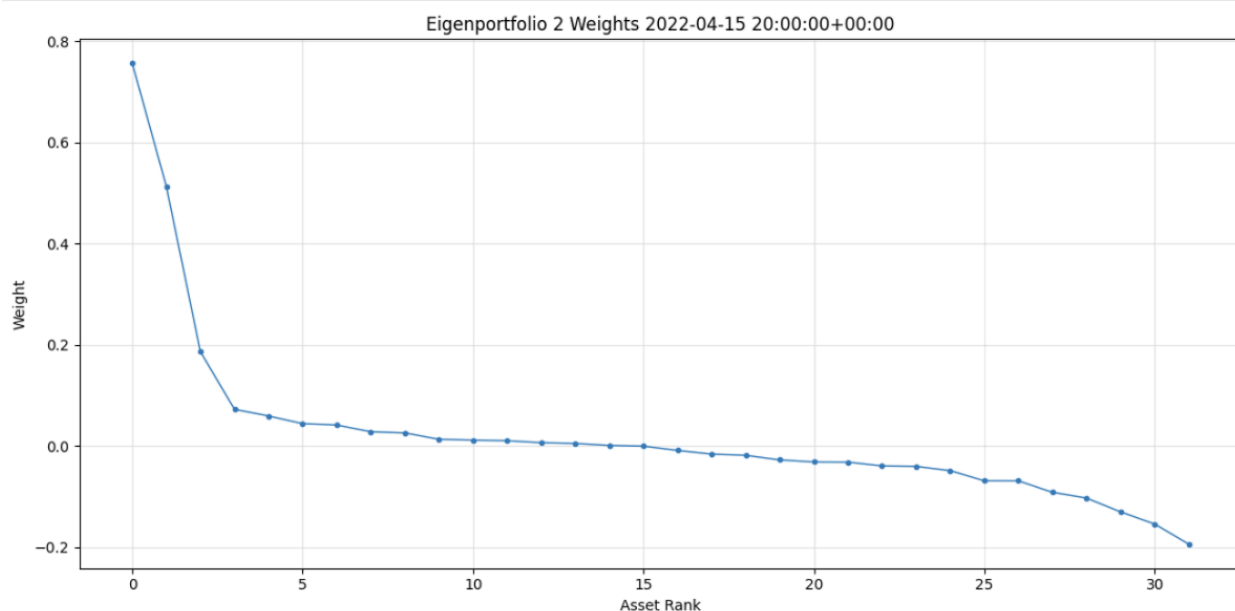


Figure 6 (*output/weights\_2\_2022-04-15.png*): Weights of the 2nd eigenportfolio at 2022-04-15T20:00:00+00:00.

### c. Task 3: S-Score Evolution

The S-score is the primary trading signal. The plots below show the evolution of the S-score for BTC and ETH.

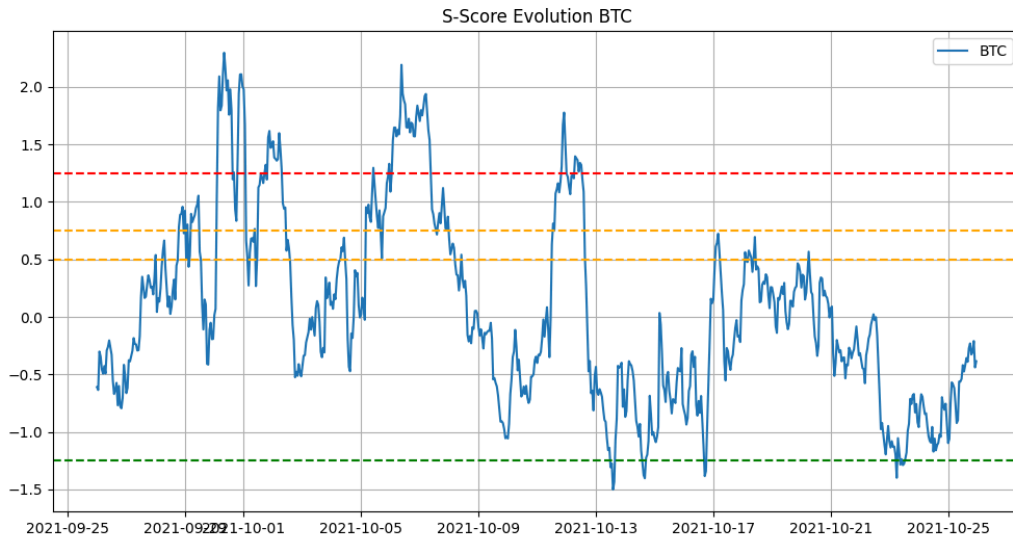


Figure 7 (*output/s\_score\_btc.png*): S-Score Evolution for BTC from 2021-09-26 00:00:00 to 2021-10-25 23:00:00.

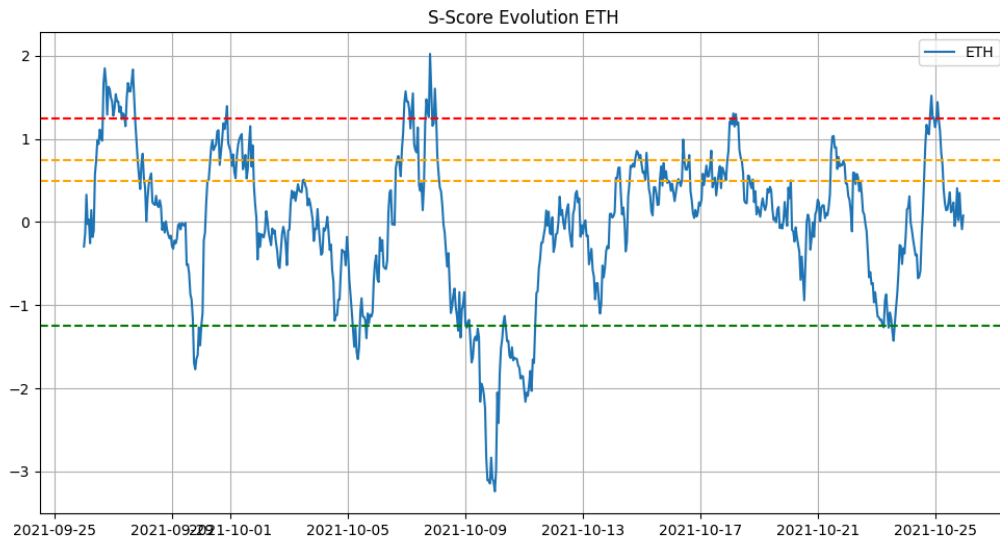


Figure 8 (*output/s\_score\_eth.png*): S-Score Evolution for ETH from 2021-09-26 00:00:00 to 2021-10-25 23:00:00.

**d. Task 4: Signal Generation and Return Distribution**

The trading signals for all 40 tokens were generated and saved to `trading\_signal.csv`. The histogram below illustrates the distribution of hourly returns, showing the strategy's skew and kurtosis profile.

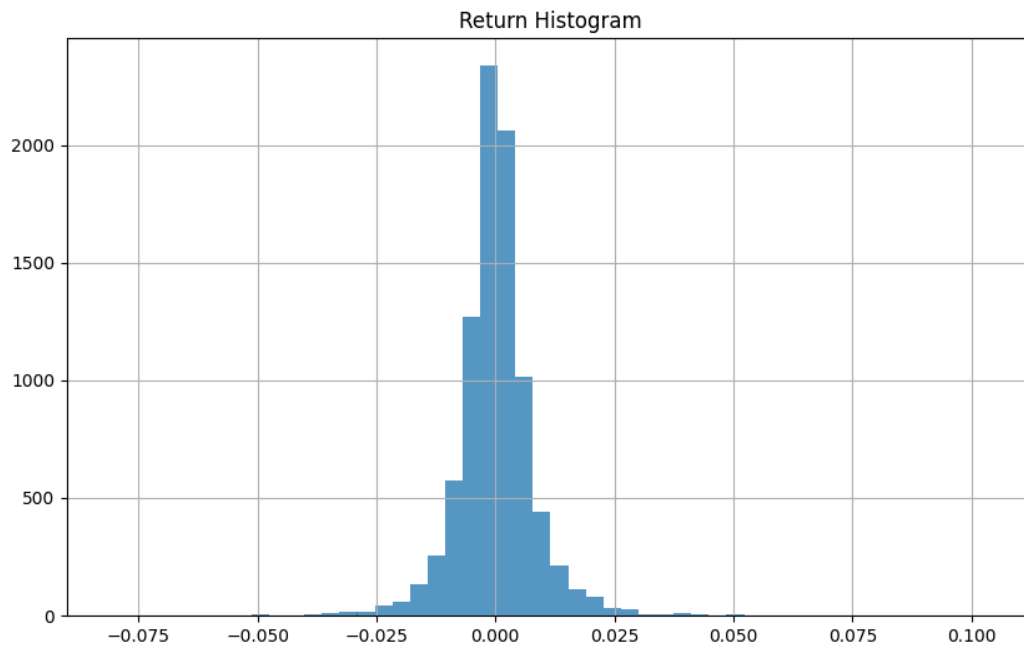


Figure 9 ([output/hist\\_return.png](#)): Histogram of Hourly Returns.

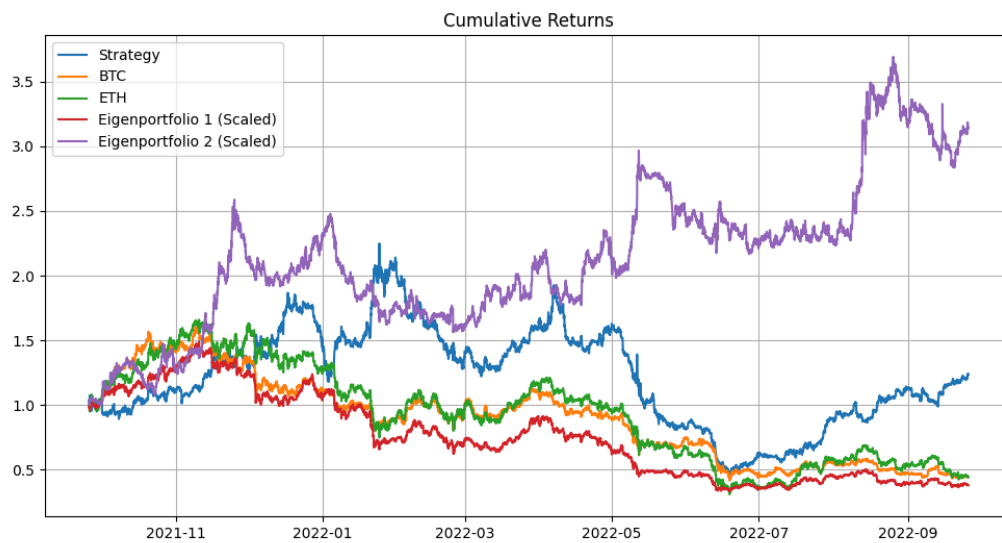


Figure 10 ([output/cumulative\\_return.png](#)): Cumulative returns of the whole implemented strategy vs benchmarks (2 eigenportfolios, BTC and ETH).

## 7. Project Result



The project successfully adapted the Avellaneda and Lee statistical arbitrage framework to the cryptocurrency market. With a rigorous Object-Oriented system and trading logic optimization for the specific constraints of the environment (high volatility + fixed unit sizing), the potentially losing strategy is converted into a profitable one with a **Sharpe Ratio of 0.6567** and a **Maximum Drawdown of -80.42%**. Although the current algorithm performs relatively well in the given data set, future improvements to improve its practicality would involve implementing fractional position sizing to better manage risk and reduce the significant drawdown attributed to capital concentration in high-priced tokens.

## 8. References

- Avellaneda, M. and Lee, J.H. (2010). "Statistical arbitrage in the US equities market". *Quantitative Finance*, 10(7), 761-782.